NTCIR-19 Kickoff Event

AgenticInstruction

Instruction Generation for Agentic Search

🔗 https://geniie-lab.github.io/ntcir/

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⚡ Motivations

- No systematic comparison of instruction designs exists for search agents.
- Prior work focuses on single search stages in isolation.
- Real search is multi-stage and iterative.
- Proposal evaluates instructions in full search sessions.

💎 Research Questions 1️⃣

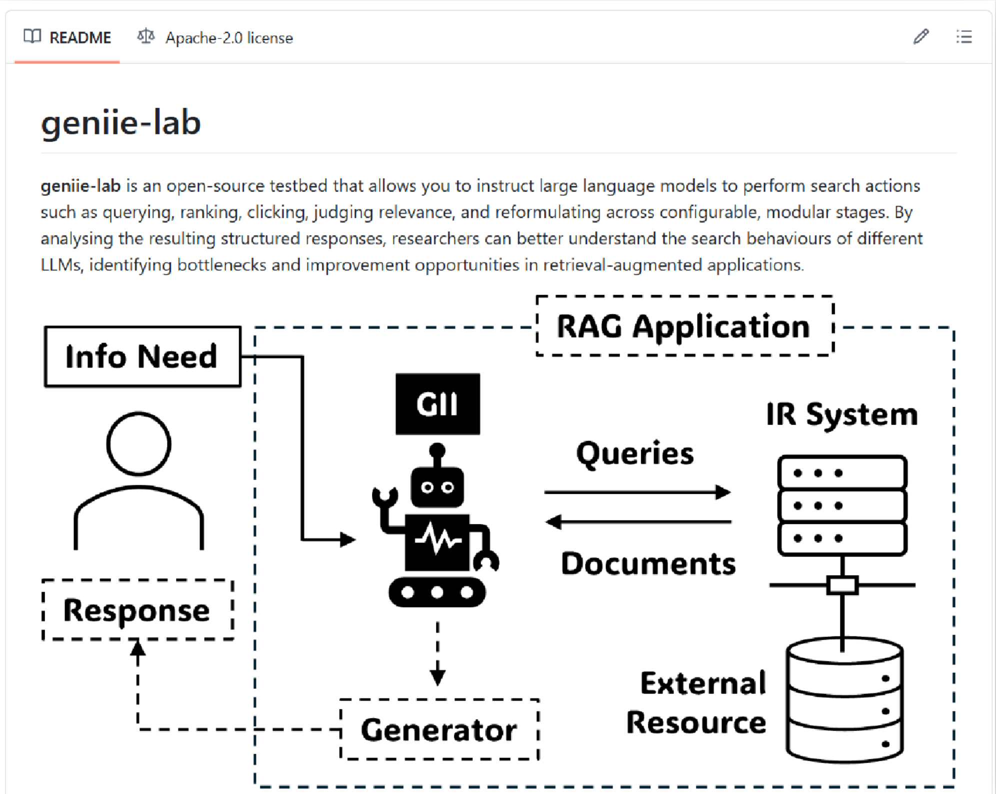
- What are the best practices for instruction design in agentic search?
- What characterises effective and robust instructions for query formulation, document selection, relevance judgment, and query reformulation?
- How does instruction performance evolve over multiple iterations in the search process?

💎 Research Questions 2️⃣

- What is the relationship between instruction design features and ranking models?
- What is the relationship between instruction design features and the choice of LLMs?
- To what extent do search topics influence instruction performance?

🥼 geniie-lab

- A Controlled Experimentation Testbed for Model Search Behaviour
- https://github.com/geniie-lab/geniie-lab
- Extension of Joho and Jose's SIGIR 2025 Paper



Instruction for Query Formulation

```
Instruction: Review the provided descriptions of
task, corpus, tool and search topic. Formulate
a search query.
============================
Task Description: {task description}
Corpus Description: {corpus description}
Search Tool Description: {tool description}
Topic Description:
- Title: {title}
- Description: {description}
- Narrative: {narrative}
```

geniie-lab Log Output


```
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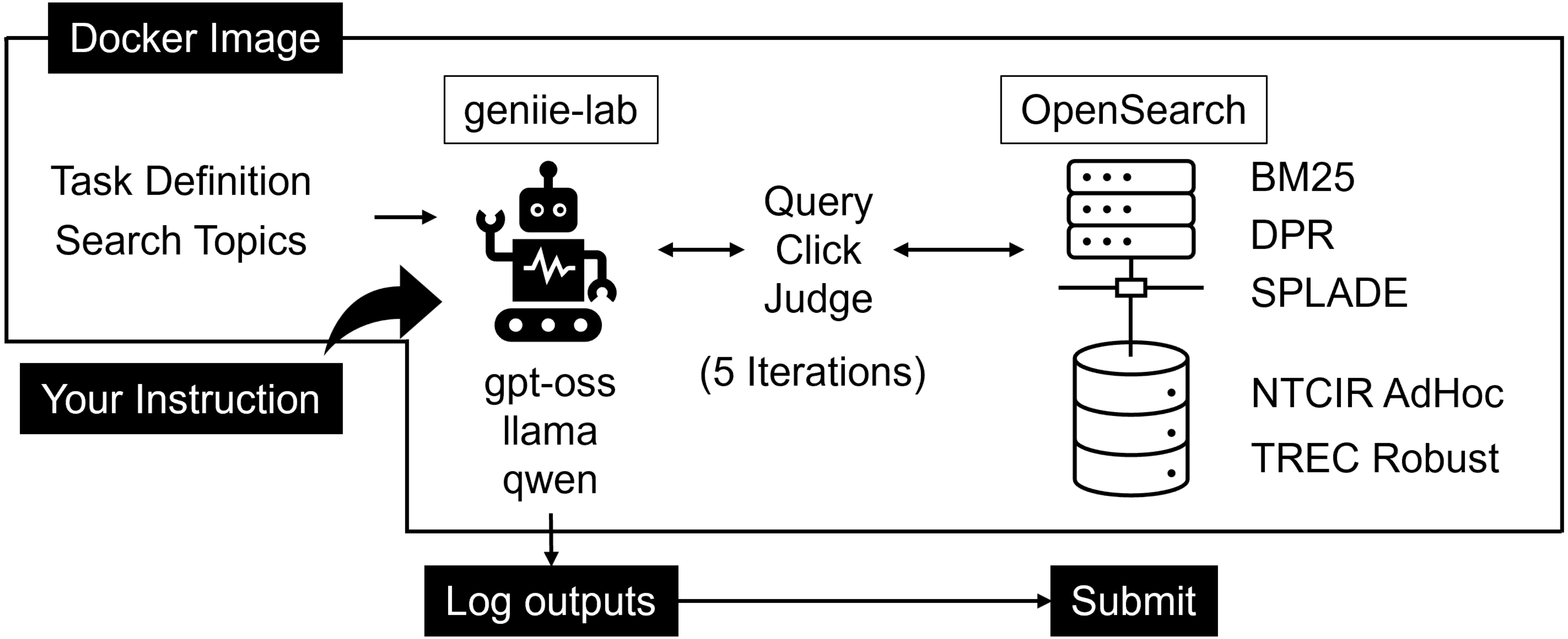
Evaluation Framework



💪 Search Task and Metrics

- Task: High-Recall Search
  - Find as many relevance documents as possible
- Metrics: Recall
  - Clicked and judged as relevant documents only
- Comparison: Recall/Token Graph
  - Y-axis: Recall, X-axis: Total Tokens

🚲 Run Types

1️⃣ Automatic Run

- Any training/human internvention on train set
- **No human intervention** on test set

💡 You must freeze your algorithm before test set.

2️⃣ Manual Run

- Any training/human internvention on train set
- **Any human intervention** on test set

Dataset

English

Train: TREC Robust 2004 Fold 1-2 (100 topics)

Test: TREC Robust 2005 (50 topics)

Corpus Genre: News articles

Japanese

Train: NTCIR-1 AdHoc (83 topics)

Test: NTCIR-2 AdHoc (50 topics)

Corpus Genre: Academic abstracts

💡 Participants must submit a user agreement form to obtain the dataset.

Tentative Schedule

Date	Activity
2025/09/03	NTCIR-19 Kick-Off Event
2025/12/07	Tutorial at SIGIR-AP 2025 (in preparation to submit)
2026/02/01	Docker image release
2026/07/01	Run submission due
2026/08/01	Evaluation results return
	Task overview release (draft)
2026/09/01	Submission due of participant papers (draft)
2026/11/01	Camera-ready participant paper due
2026/12/08-10	NTCIR-19 Conference

🤝 Stay Tuned

1. Follow `@agenticinstruct` on X for further announcements.
   - Round-table online discussion for task design
   - Tutorial on `geniie-lab`
2. Join `Discord` server for QA/Chat. `https://discord.gg/zvXkNKtEGa`
3. Email us for other enquiries.
   - `agenticinstruction-org at googlegroups dot com`
4. **WANTED:** Task Co-organiser